

Qatar-2b

R-0932

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180501 · created 2026-07-29T11:26:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458239.66613 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.221 +/- 0.066
Transit depth (Rp/Rs)^2	4.89 +/- 2.91 [%]
Orbital Inclination (inc)	84.23 +/- 1.73
Airmass coefficient 1 (a1)	0.75 +/- 0.29
Airmass coefficient 2 (a2)	0.21 +/- 0.51
Scatter in the residuals of the lightcurve fit is	36.3 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	8.43
Transit Duration (day)	0.0511 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.221 $\pm$ 0.066 vs 0.16327 (deviation 0.87 σ)
Duration fit vs published	0.0511 d vs 0.0767 d (deviation 4.19 σ)
Mid-time O-C	-1.3 min
Depth SNR	3.3
Residual scatter	36.3 %

Light curve

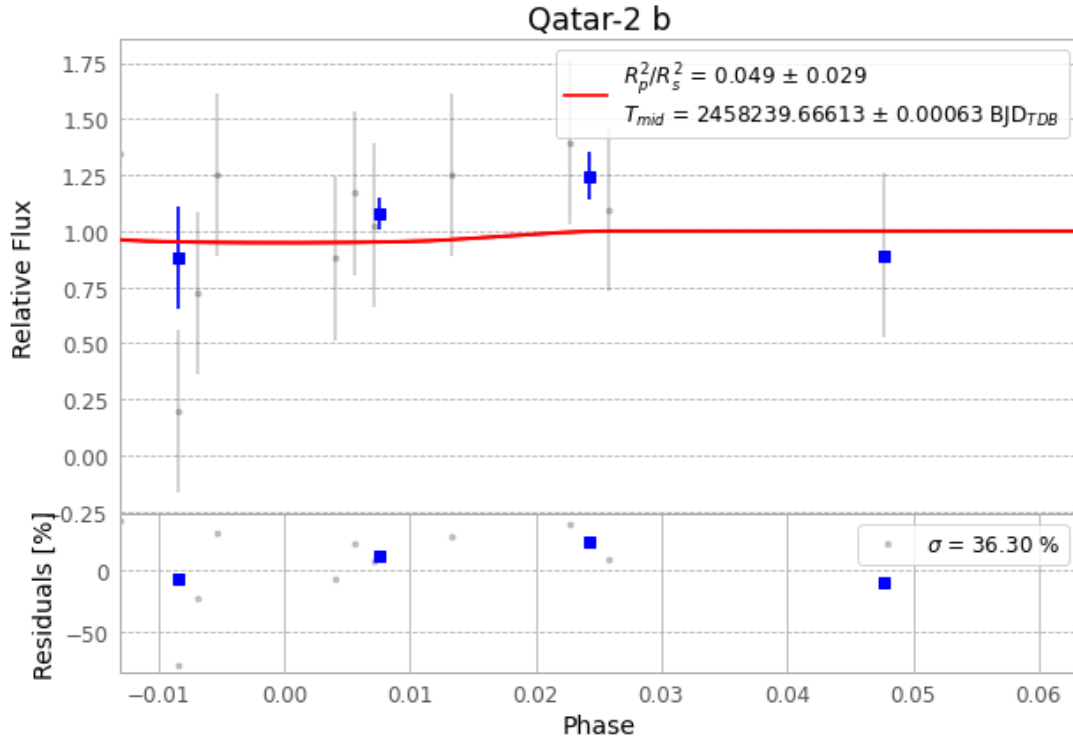


Figure 1 — FinalLightCurve_Qatar-2 b_2018-04-30.png (SHA-256 45a21a15b22e7c6c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 477], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f4480172843d172a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

53 FITS frames (35 MB), Qatar-2180501032412.FITS → Qatar-2180501060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180501.log (95797658f55b...), FinalLightCurve_Qatar-2 b_2018-04-30.png (45a21a15b22e...), FinalLightCurve_Qatar-2 b_2018-04-30.csv (68720daecddf...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:26Z.